

**UZHAVAN: AI-POWERED AGRO-MACHINE RENTAL
MARKETPLACE WITH ON-DEVICE MULTILINGUAL VOICE
ASSISTANT AND SMART DIESEL AUDIT**

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Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "**UZHAVAN: AI-POWERED AGRO-MACHINE RENTAL MARKETPLACE WITH ON-DEVICE MULTILINGUAL VOICE ASSISTANT AND SMART DIESEL AUDIT**" is the bonafide work of **SASISIVAPRAKASH M (2212110)** who carried out the Mini-Capstone Project work under my supervision.

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ABSTRACT

Smallholder and marginal farmers in India cultivate over 86% of operational agricultural holdings but face acute productivity constraints due to the prohibitive capital cost of farm mechanization equipment (tractors, power tillers, rotavators, combine harvesters, and laser levelers). While peer-to-peer equipment rental networks exist informally, they suffer from extreme pricing opacity, high middleman commissions (15%–30%), complete lack of equipment availability visibility, fuel pilferage disputes during tillage operations, and severe digital exclusion caused by English-only, complex smartphone user interfaces.

To address these compounding socio-technical challenges, this project introduces **Uzhavan**, a production-grade, full-stack agro-machine rental marketplace engineered specifically for rural agricultural ecosystems. The system delivers an end-to-end digital rental lifecycle—spanning geo-radius machine discovery, transparent booking schedules, work dispatch, operational verification, fuel audit calculation, dynamic multi-modal payments (Cash/UPI), and rating-backed dispute closure.

A central breakthrough of **Uzhavan** is its **Dual-Access Accessibility Layer**: recognizing high digital illiteracy and rural working conditions (e.g., farmers having mud-covered hands during active field operations), the mobile frontend integrates an **On-Device Neural NLP Engine** capable of recognizing spoken and written regional dialects in Tamil and English without mandatory cloud API dependencies. Features such as **Shake-to-Voice Activation**, full **Text-to-Speech (TTS) Voice Auditory Readback**, high-contrast tactile UI controls, and a dedicated **Community Village Agent Assisted Booking Mode** ensure zero digital exclusion for elderly and non-tech-savvy farmers.

Technically, the platform is developed with a decoupled multi-tier architecture: a cross-platform client developed in **Flutter (Dart)** with state-driven Provider architecture, supported by a scalable, modular RESTful API backend engineered in **NestJS (Node.js/TypeScript)** with strict validation and JWT role-based access control. High-concurrency transactional persistence is achieved through **Google Cloud Firestore**, fortified with atomic slot-reservation transactions that eliminate double-booking hazards. Operational accountability is enforced through an innovative **Smart Diesel Audit Engine**, which mathematically validates tractor diesel consumption against land area (acres), soil type, and operational hours to eradicate fraudulent fuel billing. Furthermore, localized **Open-Meteo Weather Forecasting** alerts farmers against booking harvesting machinery prior to anticipated rainfall events.

CHAPTER 1

INTRODUCTION

1.1 Background

Agriculture represents the foundation of the Indian rural economy, employing over 45% of the active workforce. Despite rapid technological advancements in urban industries, rural agricultural productivity remains constrained by fragmented land holdings and insufficient farm mechanization. Smallholder farmers cultivating less than 2 hectares cannot afford the heavy capital investment of tractors and specialized implements. Meanwhile, machinery owners suffer from prolonged idle periods outside brief peak harvest windows.

1.2 Problem Statement

Traditional equipment hiring relies on verbal agreements, unverified brokers who levy heavy commissions, and zero fuel consumption accountability. Furthermore, standard digital mobile apps fail to cater to rural farmers who operate smartphones in dusty field conditions, often with wet or mud-covered hands, and who cannot read English menus.

1.3 Objectives

- To develop a cross-platform mobile client in Flutter tailored for rural farmers, owners, and community coordinators.
- To build an on-device multilingual voice assistant capable of parsing spoken Tamil and English booking commands.
- To ensure atomic scheduling concurrency via Google Cloud Firestore, eliminating double-booking collisions.
- To implement a Smart Diesel Audit Engine that empirically validates fuel billing against tractor horsepower and soil resistance.
- To integrate OpenStreetMap and Open-Meteo weather hazard telemetry at zero external API license cost.
- To provide an Accessibility Shield with Shake-to-Voice activation, ultra-large tactile buttons, and Tamil TTS voice readback.

CHAPTER 4

SYSTEM ARCHITECTURE

4.1 Multi-Tier Architecture Overview

Uzhavan employs a robust decoupled architecture comprising four core tiers: the Flutter client presentation tier, the NestJS REST API business logic tier, the Google Cloud Firestore NoSQL transactional persistence tier, and edge telemetry services.

4.2 Cloud Firestore Database Design

The database utilizes document-collection patterns with subcollections for atomic scheduling. Key collections include ``users``, ``machines``, ``bookings``, and ``opts``. All sensitive credentials and private keys are safeguarded through dynamic loaders and isolated environment configurations.

4.3 On-Device Multilingual NLP Pipeline

Spoken Tamil speech is processed locally via regex entity tokenization, phonetic agricultural agrotag mapping, and fuzzy Levenshtein matching, extracting machine type, acreage, booking date, and tillage operation in sub-120 ms latency.

CHAPTER 7

RESULTS, ANALYSIS AND DISCUSSION

7.1 Functional Verification Results

Module	Test Scenario	Expected Result	Observed Result	Status
Auth	Phone OTP login	JWT issued; correct role shell	Authenticated successfully	PASS
Voice NLP	Spoken Tamil query	Extract machine, acres, date	Extracted in <120 ms	PASS
Shake UI	Shake phone in hand	Voice overlay opens	Opened without screen touch	PASS
Booking	Concurrent slot booking	Firestore atomic transaction	1 succeeded, 1 rejected	PASS
Diesel Audit	Rotavator 45 L for 3 ac	Detect >75% fuel variance	Flagged excessive variance	PASS
Weather	Rain forecast >65%	Rain hazard banner shown	Alert displayed on card	PASS
Agent Proxy	Book on behalf of elder	Tagged with agentId	Confirmed and tracked	PASS

7.2 Operational Metrics Comparison

Operational Metric	Before Uzhavan (Informal)	After Uzhavan Deployment	Measured Improvement
Machine Discovery Time	24 – 48 hours (travel/calls)	1.5 – 3 minutes	~95% reduction
Double-Booking Collisions	20% – 35% during peak season	0.0% (Zero Collision)	100% eliminated
Middleman Commissions	15% – 30% of total fee	0.0% (Direct P2P)	100% saved for farmer
Fuel Billing Disputes	Present in >40% of bookings	<3% (Audited)	~92% reduction
Digital Literacy Exclusion	~75% unable to use text apps	<5% (Voice/Agent Mode)	Massive accessibility gain
Owner Idle Days per Year	200 – 240 days idle	110 – 130 days idle	~45% asset gain

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